

RSM — Just-Math Chain (v9.2)

The explicit realization: coordinates, algebra, theorems. Layering: the implicit skeleton (frames, centers, coinherence, the engine) is affine — straight/curved only, no metric; the explicit layer renders it and states its own measure. [verified] marks machine-checked computation. [rendering] marks a result true of a presentation and carrying no structural load. Not sealed.

Cross-references name their item and give the number in parentheses; the name is authoritative.

Notation

- **P₀** — the total paradox: the master pair (distinction/indistinction) at equality and cancellation — not indistinction, which is one side; P names Paradox. Presenting faces |0| (collapse via emptiness) and |1| (collapse via fullness; the bars guard the reading — never the numeral, never 1_n). P₀ and P_n are one species at two scopes: P_n framed and load-borne (met at resolution), P₀ unframed and load-bearerless. An ontological-logical impossibility.
- **R_n, O_n, X_n, Y_n, G_n, B_n, P_n** — frame; origin; the two modes (magnitudes); the gradient $X_n \cdot Y_n = 1_n$; the balance $X_n = Y_n$; the joint $G_n \cap B_n$ at $(\sqrt{1_n}, \sqrt{1_n})$ [rendering coordinates].
- **σ_B** — the mode-swap $X \leftrightarrow Y$; on G it is the inversion $x \mapsto 1_n/x$, fixed point the seat; in split coordinates it fixes a and negates b, fixed set $b = 0$, the balance. **σ_B is the structural involution.**
- **v** — componentwise negation, frame-internal fixed point O_n. **v is chart.** A sign is an indexed comparison, not a second obtainer.
- Balance status: met-at-resolution, unmet-absolutely.

The reciprocity joint

1. **Co-obtaining (P2).** Neither mode reaches a terminal while the other obtains.
2. **Exchange symmetry.** The law is symmetric under $X \leftrightarrow Y$ — the structural symmetry.
3. **Class selection [verified].** Additive conservation fails twice: it admits $X = 0$ (a terminal as an attainable value) and it caps both modes (contradicting P1's vastness clause).

Multiplicative conservation does neither: $XY = 1_n$ admits no zero and no ceiling. Both of P_0 's faces are excluded here, at the gradient's two ends, by the law's own shape.

4. **Uniqueness up to reparametrization.** Any terminal-excluding symmetric law is a monotone reparametrization of the conserved product; the product form is the law in the modes' native magnitudes, and the native coordinates are fixed by the impossibility itself — they are the ones in which P_0 's faces sit at *nothing* and *without limit*. The chart is anchored by the very thing it excludes. 1_n is the conserved product, and the unit is the frame's reality. *[the native-magnitudes step: open]*

The frame, the unit, the floor

1_n is the constitutive magnitude and the minimal unit in one claim: any distinction beneath it is, by the Corollary, frame $n+1$ ("sub- 1_n " is unit-relative typing, not size — cross-frame magnitude comparison is undefined). The floor is the Definition's own boundary; no floor postulate exists. 0_n is a limit-approach, never a value. *[derived given the joint]*

The one law and its two slices

Over $\mathbb{C}$ with $z = a + bi$, one equation $Q(z) = 1_n$ carries two real slices:

- $Q_i = a^2 + b^2$ — the circle slice.
- $Q_j = a^2 - b^2$ — the gradient slice, with split unit j , $j^2 = +1$; in split coordinates $a = (X+Y)/2$, $b = (X-Y)/2$, the gradient $XY = 1_n$ is exactly $Q_j = 1_n$.

For magnitudes X, Y the sum a is positive, and **the mode-expressible domain is the branch $a \geq \sqrt{1_n}$ — the Amplitude Floor**. The branch is what the modes express; the remainder of the circle is the chart's continuation of it. The two slices meet tangentially at the seat and only there.

The circle slice. As a complete object the circle requires $a < 0$ and is therefore [rendering] — a legitimate indexed continuation. Its content at and near the seat, where it is mode-expressible, stands.

Scale and lineage, split. Within a frame, scale is motion along G : the split action $e^{\{j\phi\}}$ — hyperbolic rotation — is the orbit of rescalings $(\lambda X, Y/\lambda)$, product invariant. The family parameter $c = XY$ names the **lineage** direction along B : $c \rightarrow 0$ child-ward, $c \rightarrow \infty$ parent-ward along a lineage, root-capped; a lineage coordinate, never a global count. The lineage direction is not scale. *[derived]*

The two non-terminations exchange, pivoting on the seat *[verified]*: the inversion $x \mapsto 1_n/x$ — σ_B on G — swaps the fine center-ward direction and the vast extension; its fixed point

is $\sqrt{1_n}$. Both non-terminations are the gradient's own ends, within the frame. This is the chain's cleanest computation of the structural involution.

Theorems

T1 — disconnection *[verified]*. In the gradient slice, the level set $Q_j = 1_n$ has two components. As topology this is unconditional. The second component is the chart's image of the first, reached by v . There is one branch, and one indexed way of writing it twice.

T2 — crossing *[verified as computation; [rendering] as claim]*. Any closed spanning path crosses the null cone $Q_j = 0$; at the sign-image seats the gradient form reads -1_n . Spanning presupposes two components to span, so the claim lives in the signed chart.

The two returns *[verified as computation; [rendering] as claim]*. Along the circle, $Q_j = 1_n \cdot \cos 2\theta$: the invariant returns at π , the position at 2π . A double cover — a fact about the circle slice.

$i^2 = -1$, two routes *[rendering]*. Elimination over the five candidate closures, and the antipodal-binding argument. Results about the signed continuation.

The v-centroid *[rendering]*. Any sign-symmetric configuration sums to the center.

The parturition map *[derived; $w = z^2$ is its rendering]*. The recursion is structural: the seat is the child's origin, $O_{(n+1)}$ reads P_n , as constitution and not site-selection. $w = z^2$ realizes it, 2-to-1 with the two preimages differing by v . The two preimages are not two things bound into one; they are **one branch compared from the child's side of the joint** — indexed comparison performed across a frame boundary. This is how a sign can be measured while being chart.

Complementarity floor *[rendering; winding-measure premise open]*. Over admissible closed spanning structures, $\sigma_a \cdot \sigma_b \geq 1_n/2$, saturated by the minimal traversal. Spanning structures live in the signed chart.

The measure

The measure is definitional-by-register: Q_i is the circle slice's own quadratic form, and distances within the rendering are read from it. Whether quadraticity is *forced* rather than adopted is open — a non-quadratic candidate satisfying the coarse constraints exists and fails only the parallelogram condition. Until quadraticity is forced, "orthogonal" is a rendering fact: the skeleton says *independent*, and perpendicularity enters with the measure, downstream. The register the measure is defined on is itself *[rendering]* as a complete object, so the measure is doubly downstream of the skeleton. *[open]*

The engine (realized)

The demand: continuity entails the balance locus; $(\sqrt{1_n}, \sqrt{1_n})$ is on the curve, and balance-at-frame-resolution obtains — equal magnitudes is not collapse. Denial A (origin): one constitutive pair per frame; n has no spare pair to grant the locus. Denial B (exactness): equality at every depth is a terminal; $P1$ denies it; beneath the floor the equality opens into sub-unit imbalance. Satisfaction as organizing-locus: in-frame, circulation (the spanning structure winds the locus; traversal realizes it as ordered passage; motion only after succession); trans-frame, generation — **the child is the pursuit of exactness continuing below the floor**, its origin the approached locus read from below, realized by the parturition map. Recursion is exactness, permanently deferred. *[derived]*

No address. The seat's coordinates are where the rendering puts the joint; the constitution is not sited there or anywhere. The standing residue of sub-unit imbalance entails the child; nothing occurs at exact balance, which never obtains. The incident-pair deficit $(1 - 1_n/x_0^2)$, rendered as perpendicularity once the measure is in force) is corroborating, never load-bearing.

Anchoring. Every frame beneath n 's unit and in relation to n is parent-seeded at the balance: "beneath 1_n " is defined only relative to the parent's unit, so a sub-unit frame is by definition one the parent seeds; a free-floating sub-unit frame is a contradiction in the typing. *[open]*

Open items

1. **The open closures** — the joint's native-magnitudes step; the measure's definitional typing; the anchoring argument.
2. **The cross-frame unit relation** $1_n \leftrightarrow 1_{(n+1)}$ — whether any quantitative relation accompanies the constitutive share (structural §7). The Bekenstein exposure waits on this.
3. **Dimension three** — two rival derivations, unadjudicated; the skeleton fixes neither.
4. **Cross-frame expression** — how a parent states any fact about a child's interior.
5. **Definition T (traversal)** — the formal object realizing ordered passage. Traversal as stated runs in the signed chart and may need restating in the mode-expressible domain.
6. **The tree-question** — sibling structure and commensurability (shared ancestry as the only common register).
7. **Branching beyond the seat.** Generation has no address, so the question is whether any relation other than the constitutive pair can be inexact in the constituting way.

Candidate, unruléd: every conjugate equality within the frame reduces to $x = 1_n/x$, hence to the same seat.

8. **Quantitative complementarity** — the winding-measure premise (force from isotropy, or name as premise); the non-star-shaped extension. A question about the rendering register.

v9.2 — the chains state the framework as currently known; the editorial record holds how it got here.